MS 204 In-class Problems

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Introduction

An important skill that you should learn in college is how to make an argument.
What is an argument?

Argument

An argument consists of

- **premises** one or more propositions (sentences that are either true or false),
- **conclusion** a proposition that is arrived at by using

logical reasoning which is a rigorous way of thinking.

Two ways of studying arguments are

rhetoric — The art of using language effectively so as to persuade or influence others. *OED*

logic — The branch of philosophy that treats of the forms of thinking in general, and more especially of inference and of scientific method. *OED*

Rhetoric is traditionally studied in the humanities.
Logic is most often studied in philosophy and mathematics.

Logical reasoning

There are three types of reasoning (that I am aware of):

1. deductive reasoning
2. abductive reasoning
3. inductive reasoning

In formal logic propositions and arguments are presented in symbolic form so that propositions can be viewed as abstract objects that can only be viewed as true or false and arguments are boiled down to their structural form.
Informal logic studies arguments expressed in natural language.

Deductive reasoning

Deductive reasoning

Deductive reasoning is the process of drawing valid inferences.

If a conclusion follows logically from its premises, then the inference is valid. In other words if the premises are true then the conclusion must also be true.
Two examples of rules of inference are

1. Syllogism
2. Modus ponens

Syllogism

Example 0.1

All men are mortal.
Socrates is a man.
Therefore, Socrates is mortal.

It has the form:
All M are P.
All S are M.
Therefore, All S are P.

This is always a valid argument.
Because the two premises are true (All men are mortal, Socrates is a man) and the form is valid, the conclusion must be true (Socrates is mortal).
A valid argument with true premises is said to be *sound*.
The following argument is *valid*, but not *sound*:

Example 0.2

All moons are made of green cheese.
All demons are moons.
Therefore, all demons are made of green cheese.

Modus ponens and modus tollens

All *modus ponens* arguments have the structure:

Modus ponens

If p, then q.
p.
Therefore q.

Example 0.3

If it is raining, then I will get wet.
It is raining.
Therefore I will get wet.

Modus ponens alter ego is

Modus tollens

If p, then q.
Not q.
Therefore not p.

Example 0.4

If it is raining, then I will get wet.
I didn't get wet.
Therefore it is not raining.

Abductive reasoning

Abductive reasoning is a form of logical inference.

It looks for the simplest and most likely conclusion from a collection of observations.

It finds the most plausible conclusion, but cannot verify it definitively.

Often called "Inference to the best explanation."

Famous examples:

The Wet Lawn: You wake up, look outside, and see the grass is wet. You assume it rained last night because that is the most common reason. (Alternative: A neighbor left their sprinkler on).

Medical Diagnosis A patient comes in with a fever, sore throat, and body aches. The doctor concludes the patient has the flu, as it is the simplest explanation for the combined symptoms.

Inductive reasoning

Is a type of reasoning where the premises support the conclusion with some degree of certainty or probability if the argument is good.

One of the most famous inductive arguments that is true is that the sun will always rise in the east.

One of the most famous inductive arguments that turned out false was "all swans are white". It was so firmly believed in the West that the Roman poet Juvenal (AD 55–128) used the phrase "*rara avis in terris nigroque simillima cygno*" ("*a bird as rare upon the earth as a black swan*") to mean that an event is impossible.

It was falsified in 1697 when Dutch explorers saw black swans in Western Australia.

Chapter 1 Section 1

Determine whether the statement describes a population or a sample.

The price of homes of all the employees at a software company.

- Population
- Sample

Determine whether the statement describes a population or a sample.

The heights of 5 out of the 32 eggplant plants at Mr. Lonardo's greenhouse.

- Population
- Sample

Identify the **population** being studied.

The number of times 10 out of 20 students on your floor order pizza in a week.

- The 20 students on your floor.
- All students who ordered pizza in a week.
- The 10 students on your floor.

Determine whether the statement describes a descriptive or inferential statistic.

A recent poll of 1443 luxury car owners in West Virginia showed that the average price of a luxury car in the U.S. is $48,900.

- Descriptive Statistic
- Inferential Statistic

Determine whether the statement describes a descriptive or inferential statistic.

The average price of a car at the new car dealership in town is \$28,400.

- Descriptive Statistic
- Inferential Statistic

Determine if the numerical value describes a population parameter or a sample statistic.

A recent poll of 2935 corporate executives showed that the average price of their cars is $27,100.

- Population Parameter
- Sample Statistic

Determine if the numerical value describes a population parameter or a sample statistic.

The average price of a house in the new subdivision is $339,000.

- Population Parameter
- Sample Statistic

Identify the sample chosen for the study.

The number of times 4 out of 37 students on your floor order take-out in a week.

- The 4 students on your floor.
- All students who ordered take-out in a week.
- The 37 students on your floor.

Chapter 1 Section 2

Types of cars people own are an example of which type of data?

- Qualitative
- Quantitative
- Inferential
- Statistic

Football jersey numbers are an example of which type of data?

- Qualitative
- Quantitative
- Inferential
- Statistic

Goals scored during a soccer game are an example of which type of data?

- Qualitative
- Quantitative
- Inferential
- Statistic

Indicate the level of measurement for the data set described.

Monthly amounts of rain in Seattle over 10 years

- Interval
- Ratio
- Ordinal
- Nominal

Indicate the level of measurement for the data set described.

Categories of hurricanes that have hit the Atlantic coast

- Interval
- Ratio
- Ordinal
- Nominal

Classify data as discrete or continuous

Lengths of time it takes for new light bulbs to burn out are an example of which type of data?

- Discrete
- Continuous
- Neither

Classify data as discrete or continuous

Types of movies people go to see are an example of which type of data?

- Discrete
- Continuous
- Neither

Classify data as discrete or continuous

The numbers of each color of jelly beans in a jar (assuming they are all whole) are an example of which type of data?

- Discrete
- Continuous
- Neither

Chapter 1 Section 4

What is response bias and how can you avoid it?*

*This webpage seems to explain each type well, but I didn't read every sentence. I mainly put the link here for attributive purposes.